Ch 5.1.1-2: Leave One Out Cross-validation

Lecture 12 - CMSE 381

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Mon, Sep 30, 2024

Announcements

Last time:

- Exam

Announcements:

- Exam 1 grades.... hopefully soon
- HW #4 Posted.
  - Changed Deadline! Due Wednesday Oct 9.

Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Resampling methods: Bootstrap	5.2	HW #4 Due Weds 10/9
17	Fri	10/11	Subset selection	6.1	
18	Mon	10/14	Shrinkage: Ridge	6.2.1	
19	Wed	10/16	Shrinkage: Lasso	6.2.2	
20	Fri	10/18	Dimension Reduction	6.3	HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		

Covered in this lecture

- Validation Set
- LOO CV

Section 1

Validation set

What's the problem?

- How well is my ML method doing? *Model Assessment* → How good a job is my model doing
- Which method is best for our data?
- How many features should I use? Which ones? *Model selection*
- What is the uncertainty in the learned parameters?

Training Error vs Testing Error

Training Error

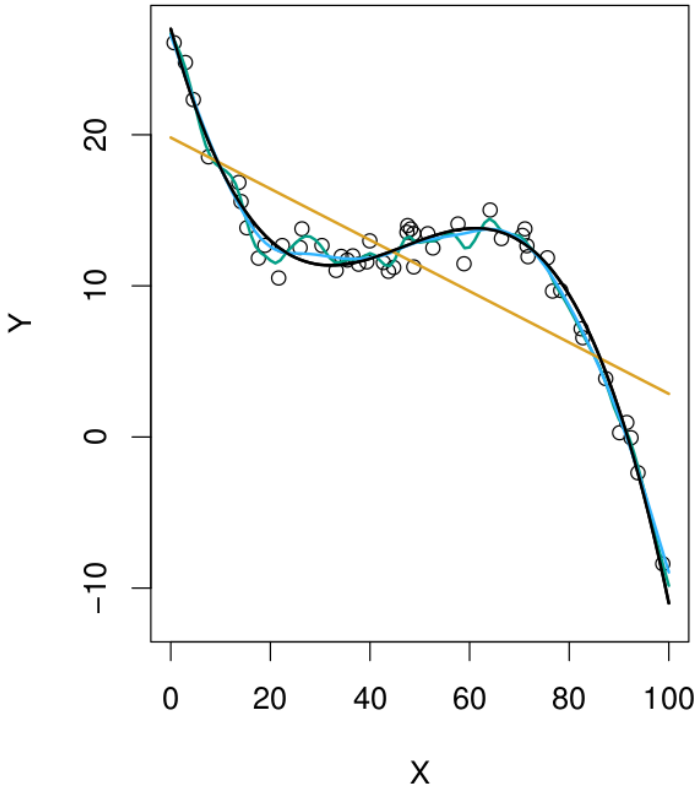
MSE
this is calculated on the training data

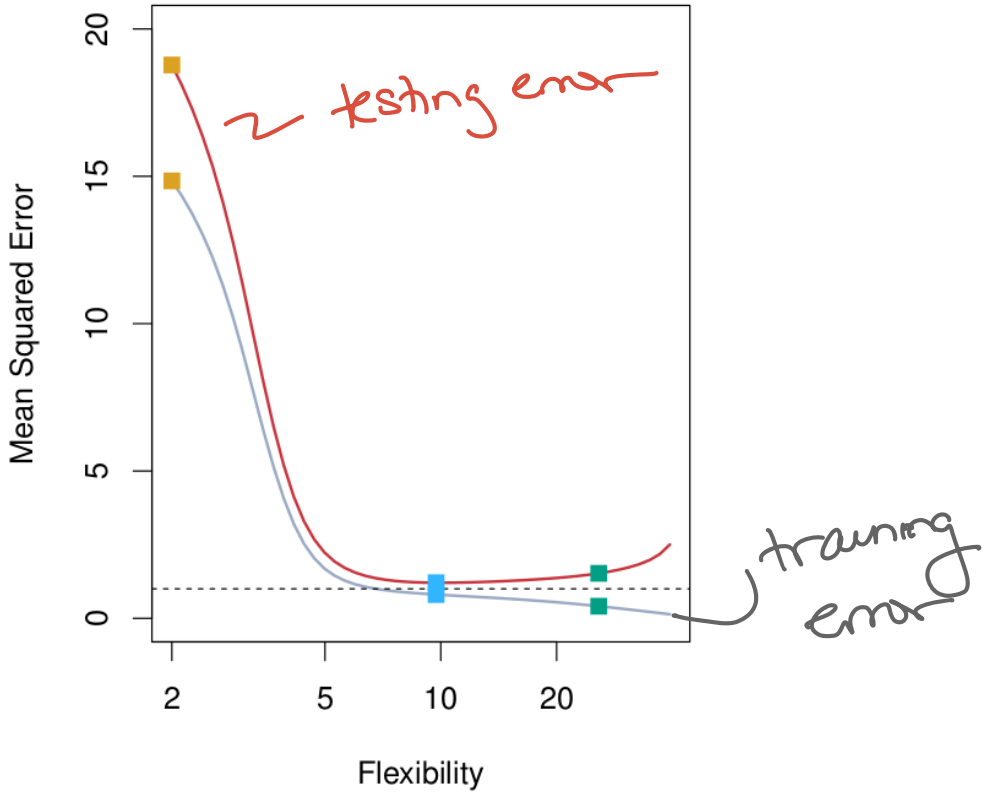
Massive under estimate of the model quality

Testing Error

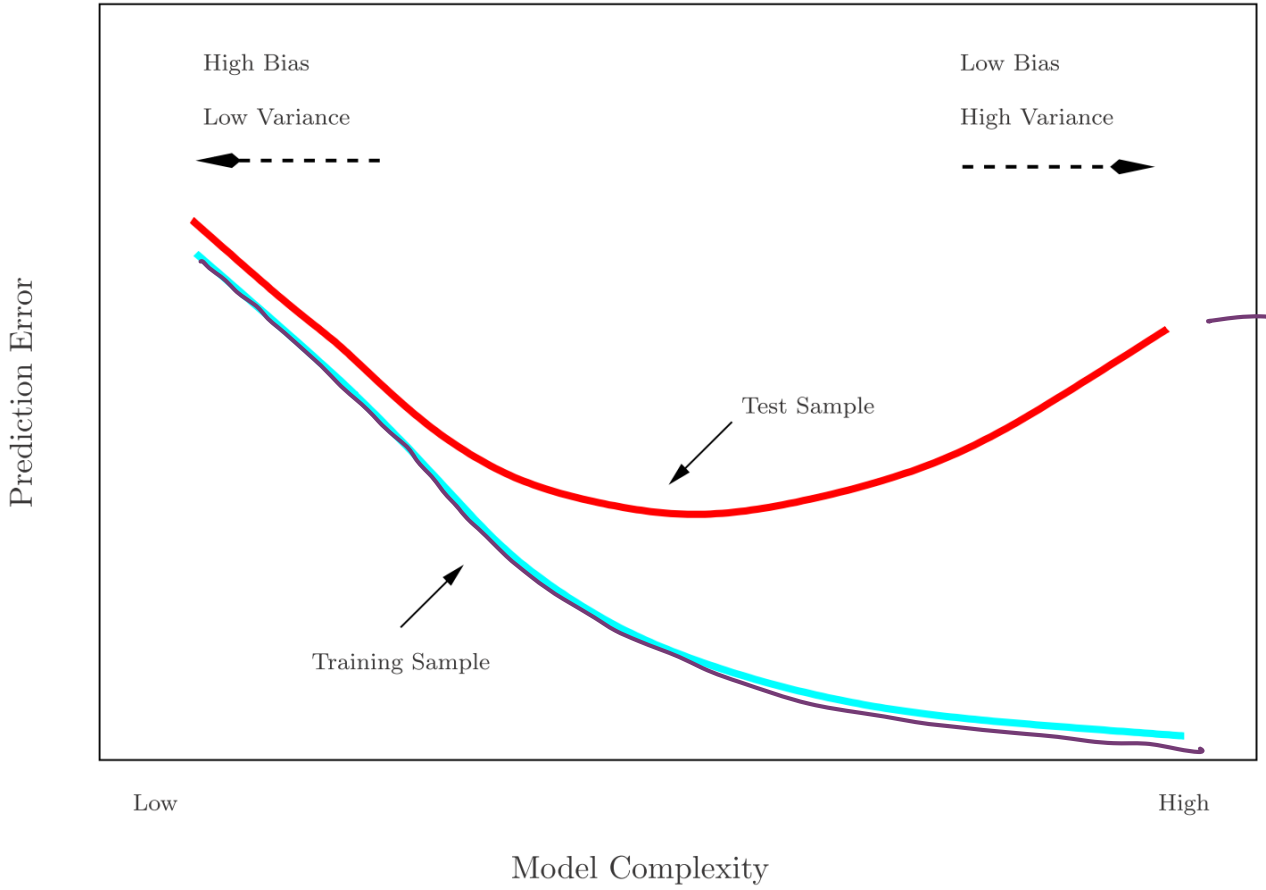
Avg error on unseen data

Throw-back Monday





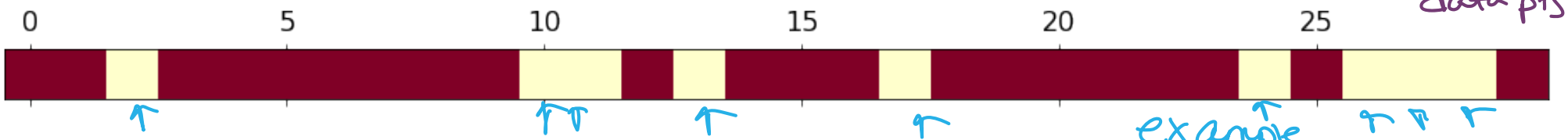
Model tradeoffs



Today estimate the test error

Validation set approach

n = example: 30 data pts



example

x = 30%

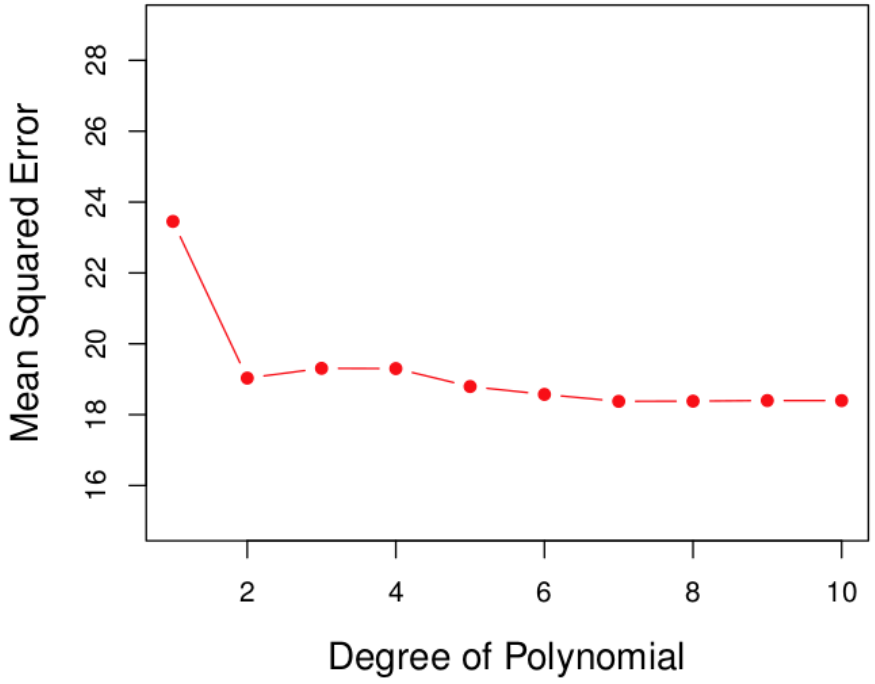
held for testing

- Divide randomly into two parts:
  - Training set
  - Validation/Hold-out/Testing set

← randomly sampled x% of data

- Fit model on training set
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

Example with the auto data



test error from validation set

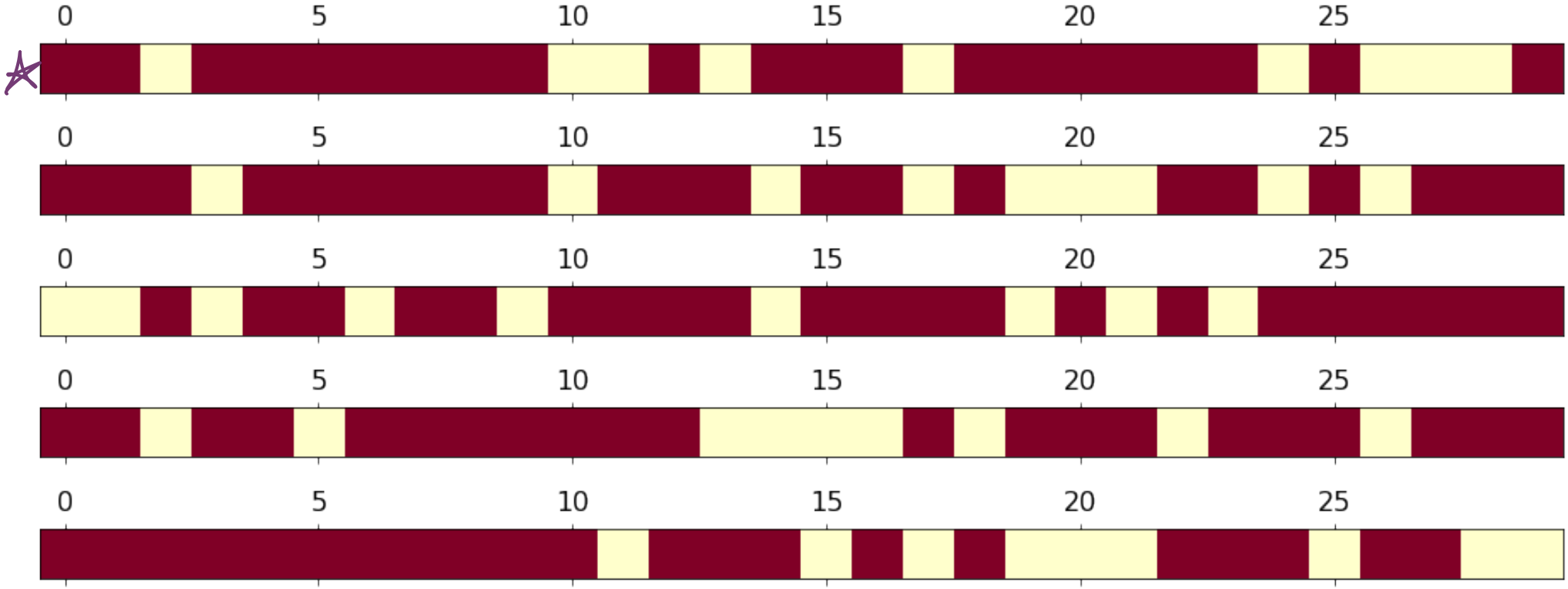
Predicting mpg using horsepower:

$$\texttt{mpg} = \beta_0 + \beta_1 \texttt{hp} + \beta_2 \texttt{hp}^2 + \cdots + \beta_p \texttt{hp}^p$$

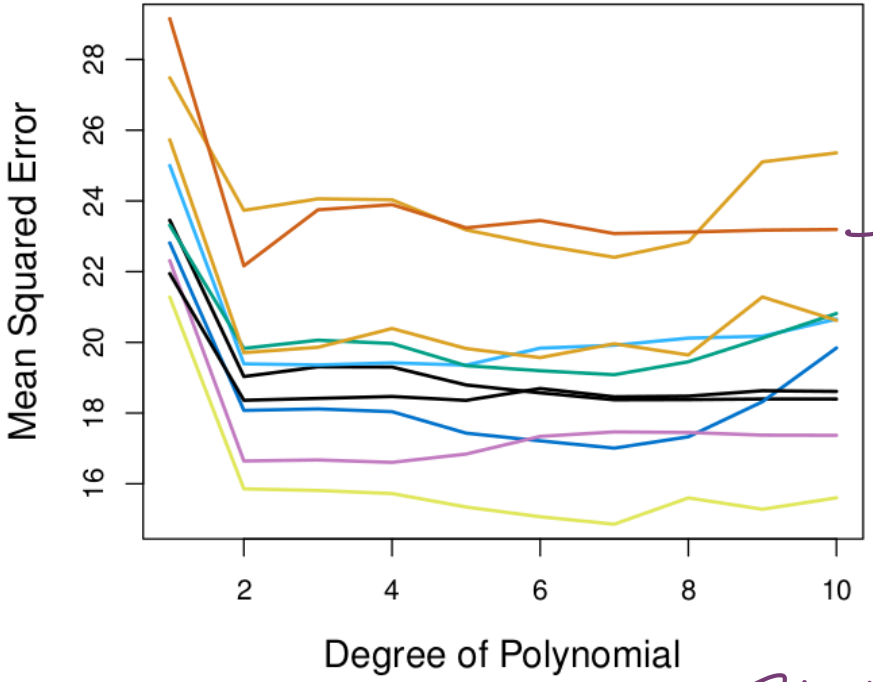
Conclusion

at least deg = 2 polynomial

Rinse and repeat



Again example with auto data



fixed line:
same training/test split
change the deg of polynomial

wide spread of options
less consensus on what the real test error is.

Stats models do worse w/ less data

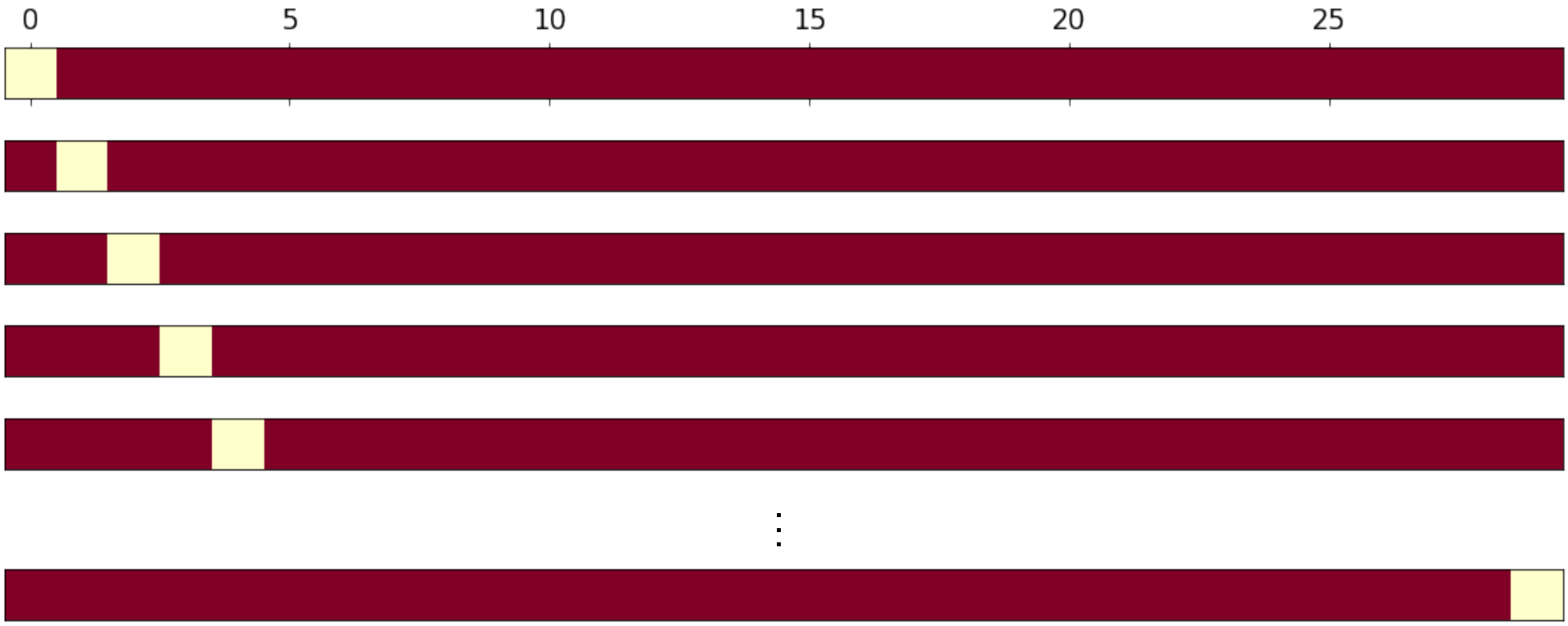
implies validation set does worse predicting test error than on the entire data set.

Coding example in jupyter notebook

Section 2

Leave-One-Out Cross-Validation (LOOCV)

The idea



The idea in mathy words

- Remove (x_1, y_1) for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1)^2$

- Remove (x_2, y_2) for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2)^2$

- Rinse and repeat

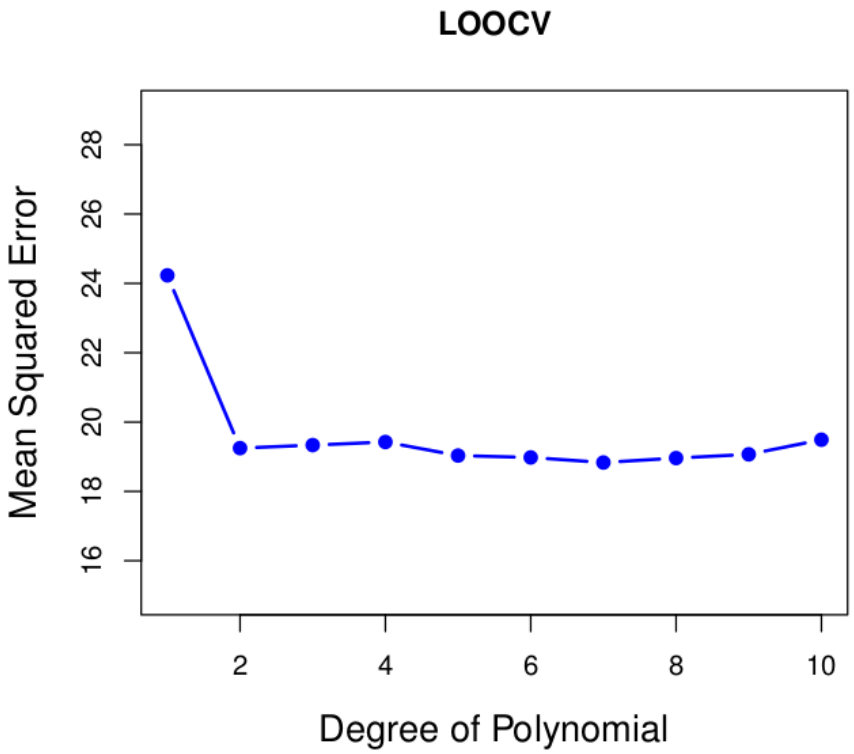
2 once for each data point

Return the score:

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

LOO cross-validation Score

Again example with auto data



There is no randomness!

Do the LOOCV coding section

LOOCV Pros and Cons

Advantages:

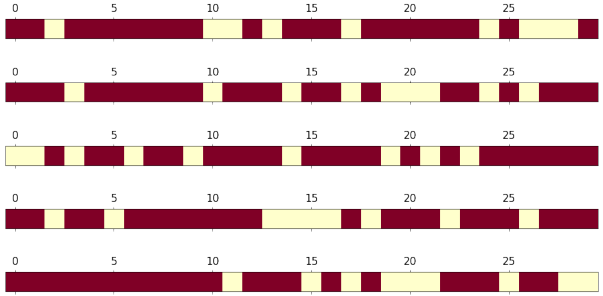
- Almost all data all the time
- Stable results:
  no randomness, same value (fixed data set) every time
- Evaluation uses all the data at some point

Disadvantages:

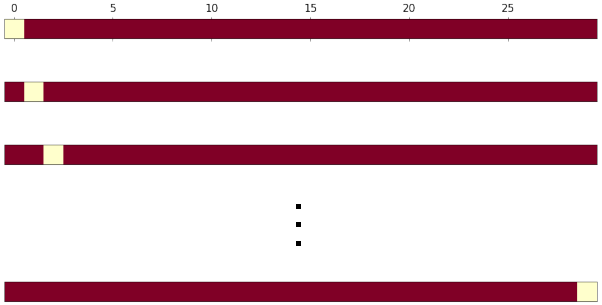
- Computationally awful
  Fit my model $n = \#$ data points times
- each of the estimates are highly correlated with each other
  data used to fit is largely the same
  $\Rightarrow$ High variance

TL;DR

Validation set



LOO-CV



LOO-CV Score

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

Next time

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